Week 4

- **Problem 1**
  Let

$$\frac{p}{q}$$

be a rational number in a reduced representation, where

$$p \in \mathbb{N}, \quad q \in \mathbb{N}, \quad 0 < p < q.$$

Suppose its decimal representation is finite.
Prove that its decimal representation terminates after *exactly two digits* after the decimal point if and only if

$$q \in \{4, 20, 25, 50, 100\}.$$

Here, "exactly two digits" means that the number can be written as a decimal with two digits after the decimal point, but cannot be written as a decimal with only one digit after the decimal point.

  - **Proof 1.1**
    Suppose $p \in \mathbb{N}, q \in \mathbb{N}, 0 < p < q$. By definition, we have $\frac{p}{q} = \frac{N}{100}$ for some $10 \leqslant N < 100$ and $10 \nmid N$.
    Then

$$100p = qN.$$

Since $\frac{p}{q}$ is a reduced representation, p and q have no common prime factors. By theorem, since the decimal representation of $\frac{p}{q}$ is finite, the only possible prime factors of q are 2 or 5. Thus, q must be a divisor of $100 = 2^2 \cdot 5^2$. We have $\operatorname{Div}_+(100) = \{1, 2, 4, 5, 10, 20, 25, 50, 100\}$.

    - ($\Longrightarrow$) Suppose $\frac{p}{q}$ has exactly two decimal digits, which means $10 \nmid N$. We prove the required set by eliminating the invalid values of q:
      - If $q = 1$, then $100p = N$. This implies $10 \mid N$, which is a contradiction.
      - If $q = 2$, then $50p = N \implies 10 \mid 50p \implies 10 \mid N$, which is a contradiction.
      - If $q = 5$, then $20p = N \implies 10 \mid 20p \implies 10 \mid N$, which is a contradiction.
      - If $q = 10$, then $10p = N \implies 10 \mid N$, which is a contradiction.
        Therefore, q cannot be 1, 2, 5, or 10. Eliminating these values from the set of all positive divisors of 100 leaves the remaining possible values:

$$q \in \{4, 20, 25, 50, 100\}.$$

    - ($\Longleftarrow$) Suppose $q \in \{4, 20, 25, 50, 100\}$. Since $\frac{p}{q}$ is a reduced representation, we know that if $2 \mid q$ then $2 \nmid p$, and if $5 \mid q$ then $5 \nmid p$. By proposition, we show that $10 \nmid N$ for each case:
      - If $q = 4$, then $100p = 4N \implies N = 25p$. Since $4 = 2^2 \mid q$, we have $2 \nmid p$, which means p is odd. Thus, $2 \nmid 25p \implies 2 \nmid N \implies 10 \nmid N$.
      - If $q = 20$, then $100p = 20N \implies N = 5p$. Since $20 = 2^2 \cdot 5 \mid q$, we have $2 \nmid p$, which means p is odd. Thus, $2 \nmid 5p \implies 2 \nmid N \implies 10 \nmid N$.
      - If $q = 25$, then $100p = 25N \implies N = 4p$. Since $25 = 5^2 \mid q$, we have $5 \nmid p$. Thus, $5 \nmid 4p \implies 5 \nmid N \implies 10 \nmid N$.
      - If $q = 50$, then $100p = 50N \implies N = 2p$. Since $50 = 2 \cdot 5^2 \mid q$, we have $5 \nmid p$. Thus, $5 \nmid 2p \implies 5 \nmid N \implies 10 \nmid N$.
      - If $q = 100$, then $100p = 100N \implies N = p$. Since $\frac{p}{100}$ is a reduced representation, we have that p cannot be a multiple of 10. Thus, $10 \nmid N$.
        In all cases, $10 \nmid N$, which means the decimal representation cannot be simplified to

have only one digit after the decimal point. Since $q \mid 100$, it terminates after at most two digits. Thus, it terminates after exactly two digits.

- **Problem 2**
  Let

$$x = \sqrt{2} + \sqrt{3}.$$

  1. Prove, without using decimal approximations, that

$$3 < x < 4.$$

  2. Prove that x is irrational.
     You may use the result that $\sqrt{2}$ is irrational. Your proof should use only the arithmetic properties of rational and real numbers and the properties of roots.

  - **Solution 2.1**
    Suppose $x = \sqrt{2} + \sqrt{3}$ then $x > 0$.

    1. We have

$$x^2 = \left(\sqrt{2} + \sqrt{3}\right)^2 = 2 + 2\sqrt{6} + 3 = 5 + 2\sqrt{6}.$$

       Since $4 < 6 < 9$, we have $\sqrt{4} < \sqrt{6} < \sqrt{9}$, that is, $2 < \sqrt{6} < 3$. Then

$$2 \times 2 + 5 < 2\sqrt{6} + 5 < 3 \times 2 + 5 \implies 9 < x^2 < 11.$$

       Thus $\sqrt{9} < \sqrt{x^2} < \sqrt{11}$. SInce $x > 0$, we have $3 < x < \sqrt{11}$. Because $\sqrt{11} < \sqrt{16} = 4,$ by transitivity of order, we obtain that $3 < x < 4$.

    2. By contradiction. Suppose $x = \sqrt{2} + \sqrt{3}$ is a rational number, that is, $x \in \mathbb{Q}$.
       Then we have

$$\begin{aligned} x = \sqrt{2} + \sqrt{3} &\implies x - \sqrt{2} = \sqrt{3} \\ &\implies \left(x - \sqrt{2}\right)^2 = \left(\sqrt{3}\right)^2 \\ &\implies x^2 - 2x\sqrt{2} + 2 = 3 \\ &\implies x^2 - 1 = 2x\sqrt{2} \\ &\implies \sqrt{2} = \frac{x^2 - 1}{2x}. \end{aligned}$$

    Since $x \neq 0$, so the denominator $2x \neq 0$. Since $x \in \mathbb{Q}$, we have that $\dfrac{x^2-1}{2x} \in \mathbb{Q}$. But $\sqrt{2} \notin \mathbb{Q}$ by proposition, which is a contradiction. Therefore, $x = \sqrt{2} + \sqrt{3}$ must be irrational.

- **Problem 3**
  Let $x > 0$. Prove that

$$\sqrt{x^{-1}} = \left(\sqrt{x}\right)^{-1}.$$

  Your proof should use the definition of the square root and the arithmetic properties of inverses rather than assuming a rule for manipulating roots.

  - **Proof 3.1**
    Suppose $x > 0$. Let $u = \sqrt{x^{-1}}$. Since $x > 0 \implies x^{-1} > 0$, definition, u is the unique positive real number satisfying

$$u^2 = x^{-1}. \quad (1)$$

    Now, let $v = \sqrt{x}$. By definition, since $x > 0$, v is the unique positive real number satisfying $v^2 = x$. Since $v > 0$, its multiplicative inverse $v^{-1} = (\sqrt{x})^{-1}$ exists and is also positive. Then by theorem, we have that

$$\left((\sqrt{x})^{-1}\right)^2 = \left(v^{-1}\right)^2 = \frac{1}{v^2} = \left(v^2\right)^{-1} = x^{-1}. \quad (2)$$

Comparing equation (1) and equation (2), both $u = \sqrt{x^{-1}}$ and $v^{-1} = (\sqrt{x})^{-1}$ are positive real numbers whose square is equal to x^{-1}. By definition, the positive square root of x is unique, that is,

$$\sqrt{x^{-1}} = (\sqrt{x})^{-1}.$$

- **Problem 4**
  Prove that for every $\varepsilon > 0$, there exists $\delta > 0$, such that

$$0 < |x-3| < \delta \implies \left|\frac{x-1}{x^2} - \frac{2}{9}\right| < \varepsilon.$$

  - **Proof 4.1**
    Suppose $\varepsilon > 0$ and let $\delta := \min\left\{1, \dfrac{36\varepsilon}{5}\right\}$. Suppose $0 < |x-3| < \delta$.
    Since $\delta \leqslant 1$, we have $|x-3| < 1$, which implies

$$-1 < x-3 < 1 \implies 2 < x < 4.$$

    Then we have

$$\left|\frac{x-1}{x^2} - \frac{2}{9}\right| = \left|\frac{9(x-1)-2x^2}{9x^2}\right| = \left|\frac{-2x^2+9x-9}{9x^2}\right| = \frac{|-(2x-3)(x-3)|}{9x^2} = |x-3| \cdot \frac{|2x-3|}{9x^2}.$$

    To bound the term $\dfrac{|2x-3|}{9x^2}$, we apply the properties of order and the triangle inequality within the restricted interval $2 < x < 4$:

    - For the numerator, since $x < 4$, we have $|2x-3| \leqslant 2|x| + 3 < 2(4) + 3 = 11$. For a tighter elementary bound without derivatives, since $2 < x < 4 \implies 4 < 2x < 8 \implies 1 < 2x-3 < 5$. Thus, we have $|2x-3| < 5$.
    - For the denominator, since $2 < x \implies x^2 > 4 \implies 9x^2 > 36 \implies \dfrac{1}{9x^2} < \dfrac{1}{36}$.

    Combining these two basic inequality bounds, we obtain the following upper bound for the fraction:

$$\frac{|2x-3|}{9x^2} < \frac{5}{36}.$$

    Since we also have $|x-3| < \delta \leqslant \dfrac{36\varepsilon}{5}$, it follows that:

$$\left|\frac{x-1}{x^2} - \frac{2}{9}\right| = |x-3| \cdot \frac{|2x-3|}{9x^2} < \left(\frac{36\varepsilon}{5}\right) \cdot \frac{5}{36} = \varepsilon.$$

    Therefore, the statement is proved.

- **Problem 5**
  Let

$$z = 1 + \sqrt{3}\,\mathrm{i}\,.$$

  (a) Compute $\overline{z}$, $|z|$, and $\dfrac{1}{z}$.
  (b) Write z in polar form $z = r\operatorname{cis}(\theta)$, where $\theta = \operatorname{Arg}(z) \in (-\pi, \pi]$.
  (c) Using de Moivre's formula, compute z^6.
  (d) Find all complex numbers w satisfying $w^3 = z$. Write your answers in polar form.
  (e) Verify directly that the three solutions from part (d) have the same modulus and that their arguments differ by $\dfrac{2\pi}{3}$.

  - **Solution 5.1**

(a) Computation of $\overline{z}$, $|z|$, and $\dfrac{1}{z}$

- By Definition 3, the conjugate of $z = 1 + \sqrt{3}\,\mathrm{i}$ is:

$$\overline{z} = 1 - \sqrt{3}\backslash\mathrm{i}.$$

- By Definition 3, the modulus of z is:

$$|z| = \sqrt{1^2 + (\sqrt{3})^2} = \sqrt{1+3} = \sqrt{4} = 2.$$

- By Theorem 3.2(2), since $z \neq 0$, the multiplicative inverse is:

$$\frac{1}{z} = \frac{\overline{z}}{|z|^2} = \frac{1 - \sqrt{3}\backslash\mathrm{i}}{2^2} = \frac{1}{4} - \frac{\sqrt{3}}{4}\backslash\mathrm{i}.$$

(b) Polar form representation of z

From part (a), we have $r = |z| = 2$. For the principal argument $\theta = \mathrm{Arg}(z)$, since z lies in the first quadrant of the complex plane because $x = 1 > 0, y = \sqrt{3} > 0$, we have

$$\tan\theta = \frac{\sqrt{3}}{1} = \sqrt{3} \implies \theta = \frac{\pi}{3}.$$

Thus $\mathrm{Arg}(z) = \dfrac{\pi}{3}$. Therefore, the polar form of z is:

$$z = 2\,\mathrm{cis}\left(\frac{\pi}{3}\right).$$

(c) Computation of z^6 using de Moivre's formula

- By Corollary,

$$\begin{aligned} z^6 &= \left(2\,\mathrm{cis}\left(\frac{\pi}{3}\right)\right)^6 \\ &= 2^6\,\mathrm{cis}\left(6 \cdot \frac{\pi}{3}\right) \\ &= 64\,\mathrm{cis}(2\pi). \end{aligned}$$

- Since $\mathrm{cis}(2\pi) = \cos(2\pi) + \mathrm{i}\sin(2\pi) = 1 + 0\,\mathrm{i} = 1$, we obtain that

$$z^6 = 64 \cdot 1 = 64.$$

(d) Find all complex numbers w satisfying $w^3 = z$:

- Let $w = \rho\,\mathrm{cis}(\varphi)$ be the polar representation of the roots. By De Moivre's formula, $w^3 = \rho^3\,\mathrm{cis}(3\varphi)$. We set $w^3 = z$:

$$\rho^3\,\mathrm{cis}(3\varphi) = 2\,\mathrm{cis}\left(\frac{\pi}{3}\right).$$

- Comparing the moduli gives $\rho^3 = 2 \implies \rho = \sqrt[3]{2}$.
- Comparing the arguments gives $3\varphi = \dfrac{\pi}{3} + 2k\pi \implies \varphi_k = \dfrac{\pi}{9} + \dfrac{2k\pi}{3}$ for $k \in \mathbb{Z}$. To obtain three distinct roots, we choose $k = 0, 1, 2$:
  - For $k = 0$: $\varphi_0 = \dfrac{\pi}{9} \implies w_0 = \sqrt[3]{2}\,\mathrm{cis}\left(\dfrac{\pi}{9}\right)$.
  - For $k = 1$: $\varphi_1 = \dfrac{\pi}{9} + \dfrac{2\pi}{3} = \dfrac{7\pi}{9} \implies w_1 = \sqrt[3]{2}\,\mathrm{cis}\left(\dfrac{7\pi}{9}\right)$.
  - For $k = 2$: $\varphi_2 = \dfrac{\pi}{9} + \dfrac{4\pi}{3} = \dfrac{13\pi}{9} \implies w_2 = \sqrt[3]{2}\,\mathrm{cis}\left(\dfrac{13\pi}{9}\right)$.
- Therefore, the three solutions in polar form are w_0, w_1, and w_2 as defined above.

(e) Verification of the modulus and argument properties:

- **Modulus verification**: From part (d), the moduli of the three solutions are $|w_0| = |w_1| = |w_2| = \rho = \sqrt[3]{2}$. Thus, they all have the exact same modulus.

- **Argument verification**: We directly compute the adjacent differences between the arguments $\varphi_0, \varphi_1, \varphi_2$:

$$\varphi_1 - \varphi_0 = \frac{7\pi}{9} - \frac{\pi}{9} = \frac{6\pi}{9} = \frac{2\pi}{3}.$$

$$\varphi_2 - \varphi_1 = \frac{13\pi}{9} - \frac{7\pi}{9} = \frac{6\pi}{9} = \frac{2\pi}{3}.$$

$$(\varphi_0 + 2\pi) - \varphi_2 = \frac{19\pi}{9} - \frac{13\pi}{9} = \frac{6\pi}{9} = \frac{2\pi}{3}.$$

  Therefore, the arguments of the three solutions differ consecutively by exactly $\frac{2\pi}{3}$.